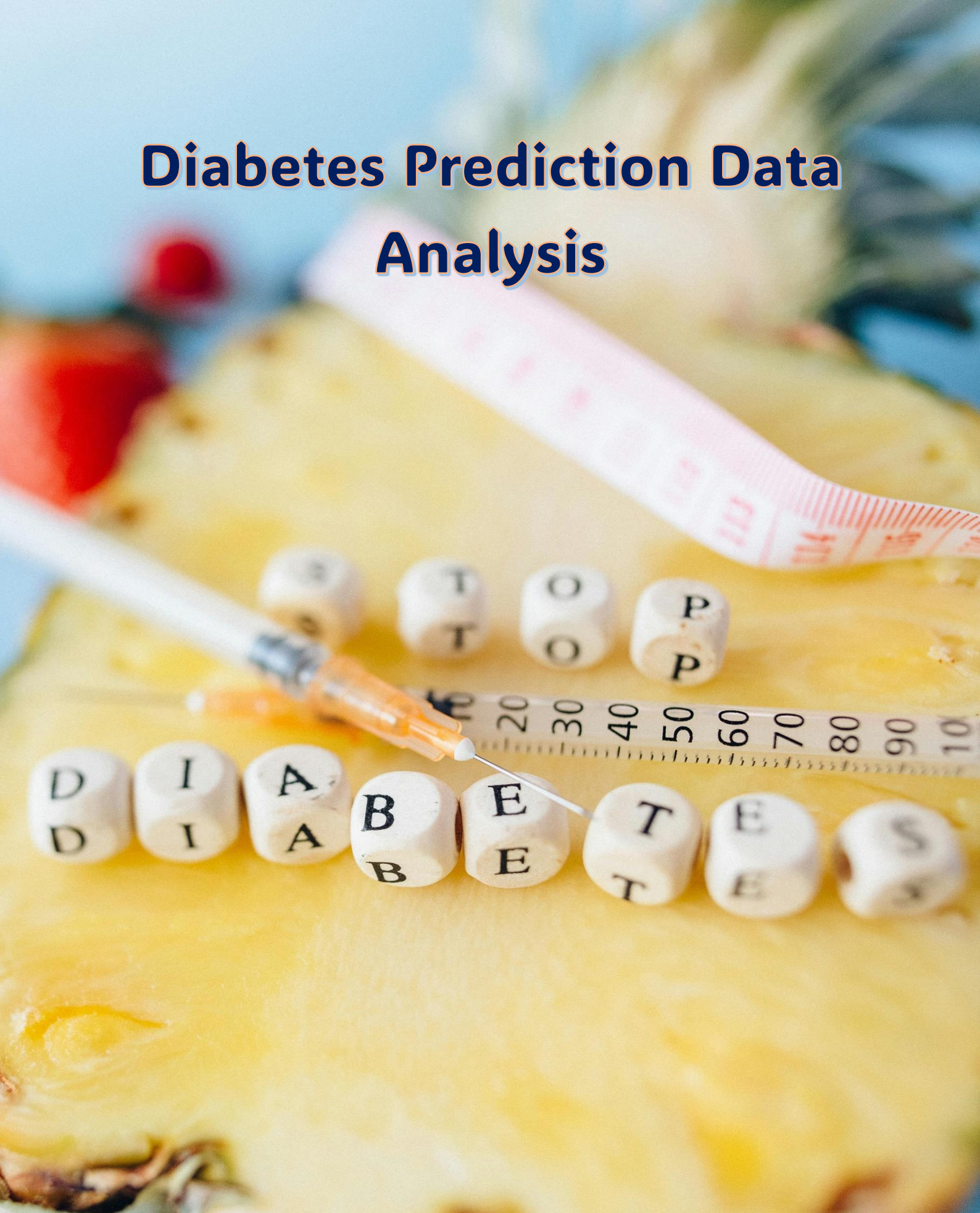


Diabetes Prediction Data Analysis



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Introduction

Diabetes is a common illness that affects many people and is a major public health issue both in the UK and throughout the world. It was estimated in 2017 that 3.7 million individuals in the UK had diabetes, which contributed to almost 700 deaths a week in the whole country [1]. As one of the main causes of disability and death in the UK, diabetes affects around 4.9 million individuals, according to the latest statistics.

With one in ten persons in the UK being diagnosed with the disease, the frequency of diabetes is concerning. It is estimated that by 2025, up to five million individuals in the UK may have diabetes if the present scenario continues [2]. Due to long-term tissue damage, diabetic patients are more likely to have medical complications such as kidney failure, eyesight loss, stroke, heart disease, and early death. Furthermore, diabetes has a significant financial impact; in the UK, the annual cost of treating diabetes is estimated to be £14 billion [3].

People who are at high risk for diabetes may be unaware of the risk factors. Therefore, recognising the most frequent diabetes risk factors is critical to lowering diabetic complications and the economic burden. According to current research, the key risk factors for diabetes are eating habits, age, family history, ethnic groupings, obesity, a sedentary lifestyle, and an earlier diagnosis of gestational diabetes. Previous research has also found that sex, pregnancy, BMI and metabolic condition have a connection to diabetes.

In this report, we will cover the dataset that we got from this website (<https://www.kaggle.com/datasets/samira1992/diabetes-intermediate-dataset/data>), where the publisher's name is Samira Shemirani, and the dataset was last updated in 2024. This data came from the National Institute of Diabetes, Digestive and Kidney Diseases. The dataset's goal is to diagnose whether or not a patient has diabetes based on various diagnostic metrics such as the patient's number of pregnancies, BMI, insulin level, age, and other factors. Several limitations were imposed on the choice of these instances from a broader database. Every single patient here is a female aged at least 21 [4].

Dataset Description:

Numerical Data	Features	Description
	Pregnancies	This column provides information on the number of times a patient was pregnant.
	Glucose	This column tells about plasma glucose concentration at 2 hours in an oral glucose tolerance test.
	BloodPressure	This column shows diastolic blood pressure (mmHg).
	SkinThickness	This column provides information on triceps skinfold thickness (mm)
	Insulin	This column provides information about 2-hour serum insulin (mu U/ml).
	BMI	This column indicates body mass index (BMI), calculated as weight in kg divided by the square of height in meters.
	DiabetesPedigreeFunction	This column represents the Diabetes Pedigree Function.
	Age	This column provides information about years.
	Outcome	This column in the dataset is the target, where 0 represents 'No' (healthy), and 1 represents 'Yes' (diabetic)

Hypothesis

Hypothesis 1: Identifying Risk Factors of Diabetes Based on Glucose, Blood Pressure, and BMI

A group of doctors and scientists is conducting research to determine the level of glucose, Blood Pressure, and BMI of a patient that puts him at a high risk of Diabetes. They want to know exactly what levels increase the risk of developing diabetes. By determining this, they may alert people early on whether they require medical assistance. This will make diabetes treatment easy and make people healthy.

Hypothesis 2: Age-Based risk of getting Diabetes.

Doctors and scientists are also examining which age groups are most vulnerable to developing diabetes. They want to know which persons in various age groups are most likely to have the disease. Once they understand, they may inform those age groups about the dangers of diabetes. This way, people in those age groups may take steps to be healthy and seek help when required.

Hypothesis 3: Predicting Diabetes Risk Based on Family History (Diabetes Pedigree Function).

Doctors and scientists are doing important research to predict the risk of getting diabetes based on family history, with a special focus on the Diabetes Pedigree Function (DPF). Their major objective is to determine the efficacy of using DPF along with other health indicators, such as the number of births and skin thickness, in predicting an individual's risk of developing diabetes.

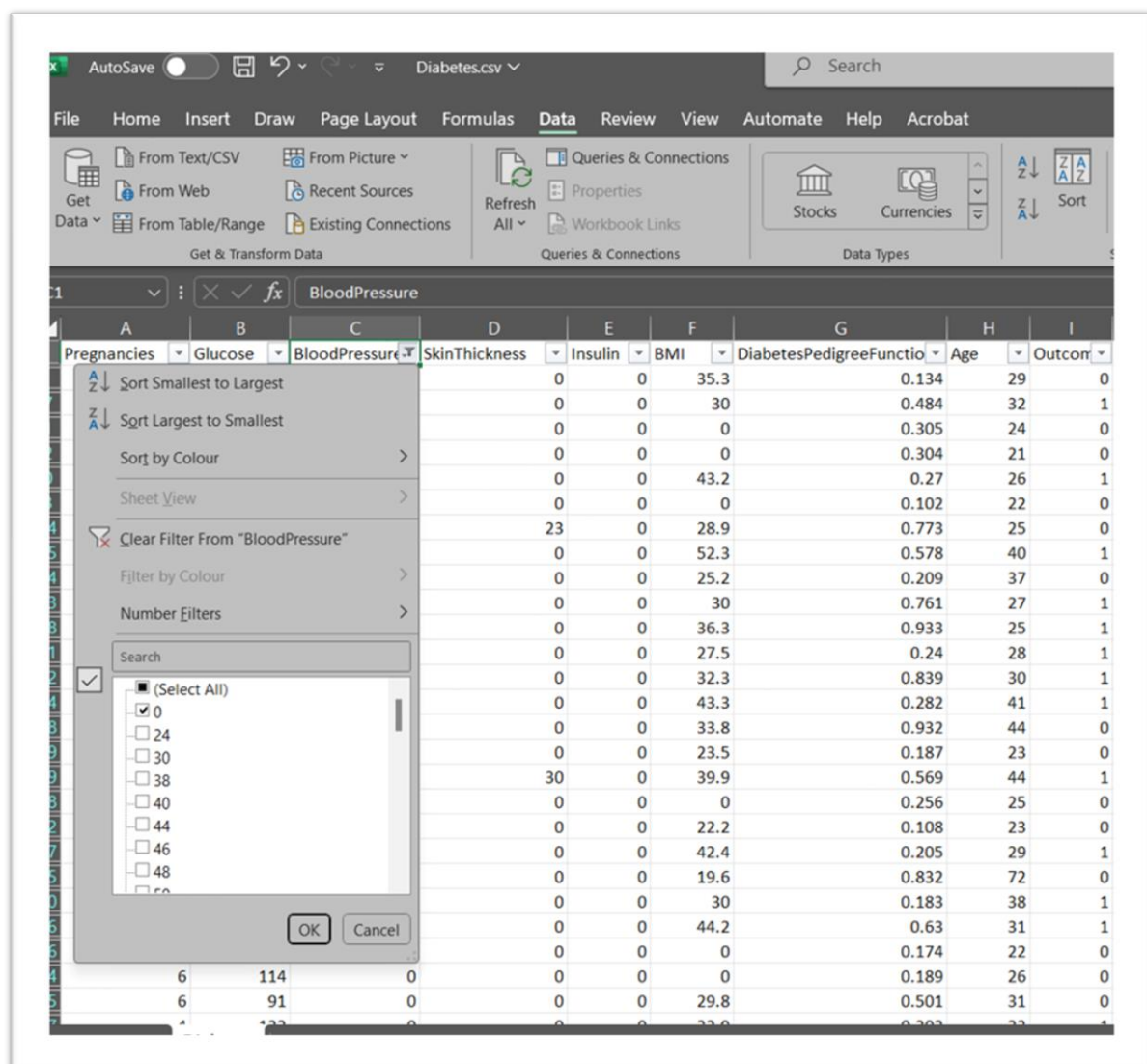
Researchers want to identify those who are at a higher risk of acquiring diabetes or who already have the disease but are unaware of it. By using this

knowledge, medical professionals may provide patients with individualised advice on adopting proper dietary habits and lifestyle changes to reduce the risk of diabetes development.

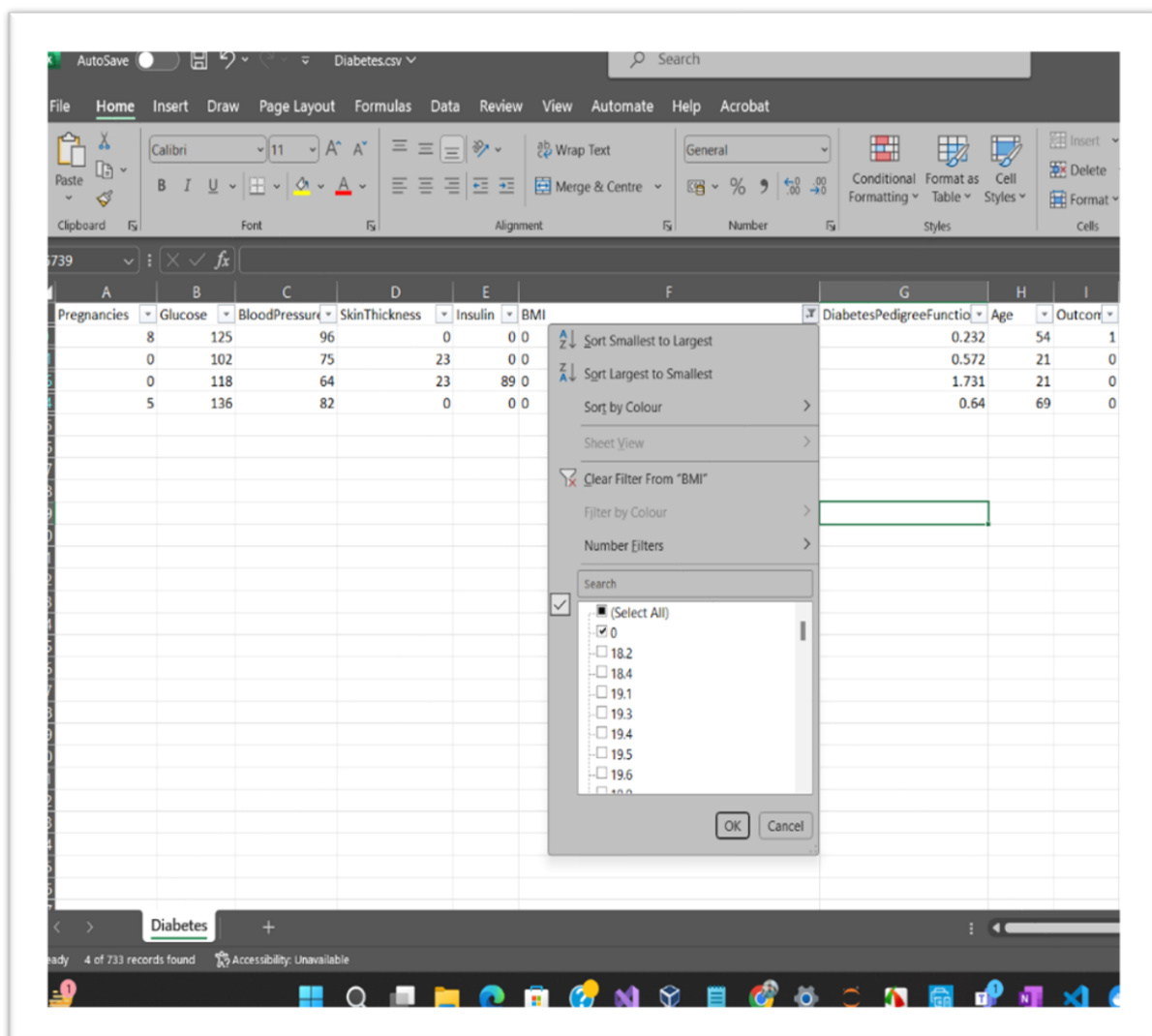
In addition, Early detection is key for those at risk of diabetes due to their genetics. It will allow doctors to act fast, manage the condition, and avoid complications.

PRE-PROCESSING

Step 1: I deleted all the records with a blood pressure of 0 because it is not possible for a living person to have a blood pressure of zero. Its shown on Picture below:

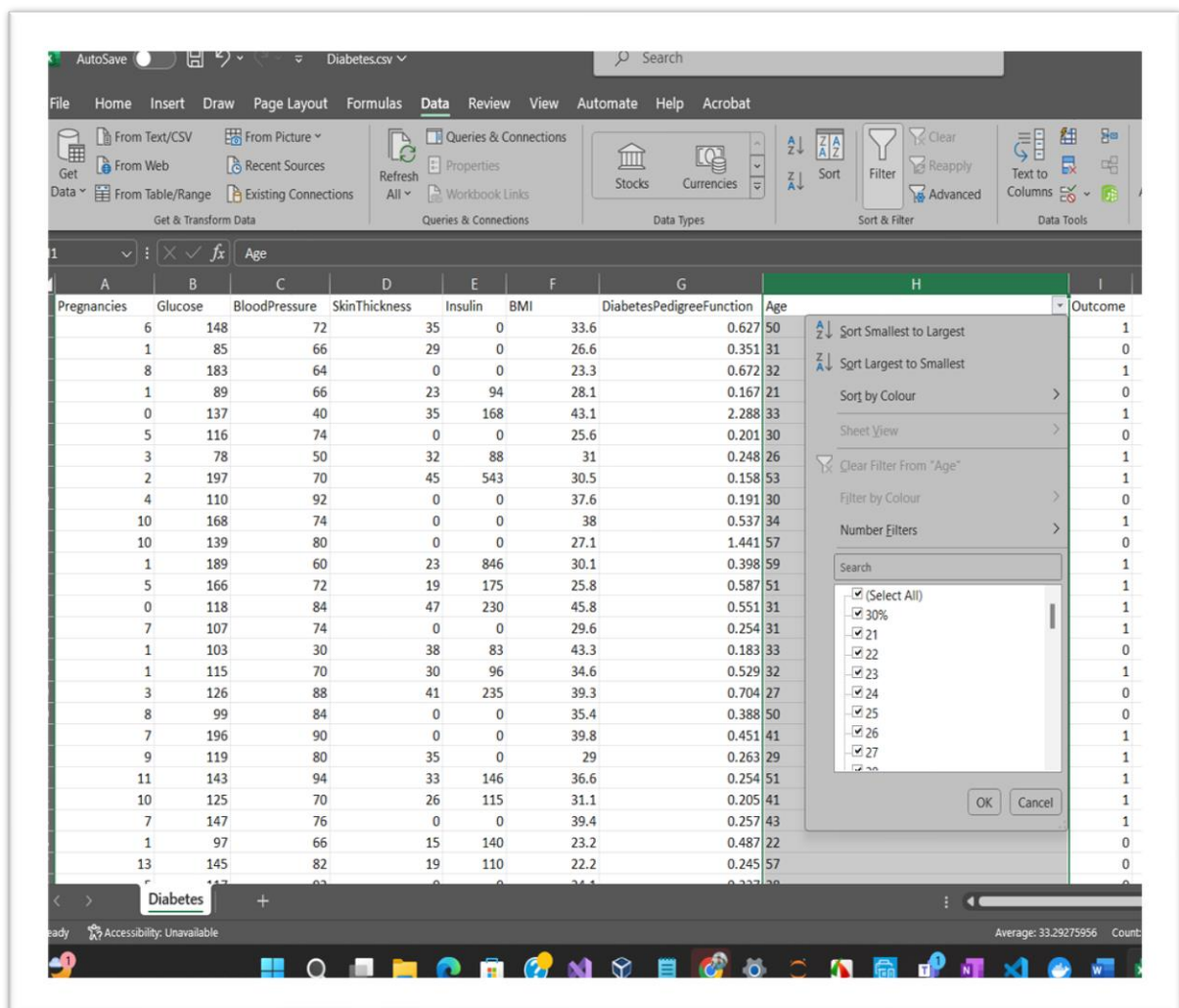


Step 2: I deleted all the records with a BMI of 0 because it is not possible for a living person to have a BMI of zero. The calculation of a person's BMI involves dividing their weight in kilograms by the square of their height in meters. The BMI of an individual would remain higher than zero even if they were severely underweight or had a very low weight as a result of disease, accident, or other extreme circumstances. A person with a BMI of zero would also appear to have no mass at all, which is not possible in the real world.



Step 3: The age column has the wrong format in one few cell. I did correct that.

Shown in picture:



Proposed Solution:

The **K-Nearest Neighbour(K-NN)** algorithm can effectively solve the problems proposed in the hypothesis. It is a powerful tool that is capable of handling both classification and regression problems. By finding the nearest neighbour within a certain area around a given point, the algorithm identifies the closest attribute to make an accurate prediction. This technique is commonly known as supervised learning and has proven to be highly effective in various applications.

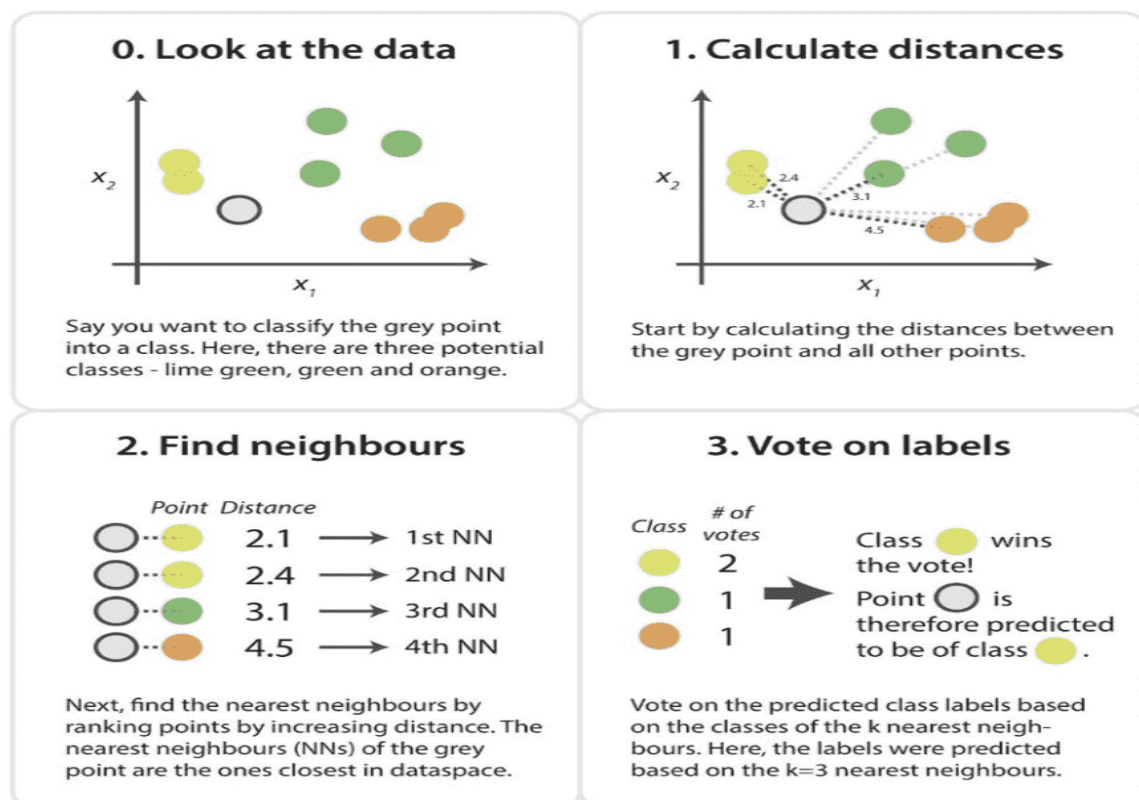


Image: K-Nearest Neighbor © medium.com

For hypothesis 1, the K-Nearest Neighbor method will use patient data such as glucose levels, blood pressure, and BMI measures. It will measure the distance with Neighbor data points. Then, it will find the k number of smallest distances. After that, it will find the majority outcome, which will be the predicted result.

For Hypothesis 2, K-Nearest Neighbours (KNN) will use patient data such as age and diabetes status to identify age-related risk variables for diabetes. The algorithm will compute the distance between each patient's age and that of nearby data points. It will then determine the k number of nearest neighbours

with comparable ages. KNN will forecast the chance of diabetes development for people of various ages by analysing the most common result among these neighbours.

For Hypothesis 3, K-Nearest Neighbours (KNN) will use patient data such as Diabetes Pedigree Function (DPF), and other health indicators to predict diabetes risk. The algorithm will determine the distance between each patient's data and that of neighbouring data points, taking into account DPF. It will then discover the k nearest neighbours with comparable characteristics. By analysing the majority result among these neighbours, KNN predicts an individual's likelihood of acquiring diabetes based on their genetic history and other health characteristics.

Reference:

- [1] Diabetes UK "Diabetes facts and stats" 2018. Available: <https://www.diabetes.org.uk/resources-news-expert-opinion/facts-and-stats>
- [2] Colman, P. G., & Donnelly, R. (2017). Diabetes in the UK 2017. Practical Diabetes International, 34(8), 380–382. doi: 10.1002/pdi.2154
- [3] Anonymous "Cost of Diabetes." 2019. Available: <https://www.diabetes.co.uk/cost-of-diabetes.html>
- [4] S. SHEMIRANI, "Diabetes - Intermediate  Dataset," 2022. Available: <https://www.kaggle.com/datasets/samira1992/diabetes-intermediate-dataset>